

Algebra Essentials



Evaluating expressions

- 1 If $x = 4$, evaluate $3x + 7$.
 - 2 If $a = 5$ and $b = 2$, evaluate $a^2 - 3b$.
 - 3 If $x = -3$, evaluate $2x^2 + 5x$.
 - 4 If $p = 6$ and $q = -2$, evaluate $\frac{p - q}{q + 6}$.
-

Solving linear equations

- 5 Solve $2x + 5 = 17$.
 - 6 Solve $5x - 8 = 27$.
 - 7 Solve $3(x - 4) = 21$.
 - 8 Solve $\frac{x}{4} + 3 = 10$.
 - 9 Solve $7x + 3 = 4x + 18$.
-

Expanding and simplifying

- 10 Expand $(x + 3)(x + 5)$.
 - 11 Expand $(x - 4)(x + 6)$.
 - 12 Expand $(2x + 1)(x - 3)$.
 - 13 Simplify $3(2x - 5) - 2(x - 4)$.
-

Factoring

- 14 Factor $x^2 + 9x + 20$.
- 15 Factor $x^2 - 2x - 15$.

16 Factor $x^2 - 49$.

17 Factor $6x + 15$.

Solving quadratic equations by factoring

18 Solve $x^2 + 5x + 6 = 0$.

19 Solve $x^2 - 7x + 12 = 0$.

20 Solve $x^2 - 16 = 0$.

21 Solve $x^2 + 2x - 15 = 0$.

Systems of two equations

22 Solve the system: $x + y = 10$, $x - y = 4$.

23 Solve the system: $2x + y = 11$, $x - y = 1$.

24 Solve the system: $x = 3y - 1$, $2x + y = 19$.

25 Solve the system: $4x - y = 10$, $2x + y = 8$.

26 Solve the system: $5x + 3y = 21$, $2x - 3y = 0$.

Laws of exponents

27 Simplify $a^5 \times a^3$.

28 Simplify $\frac{x^9}{x^4}$.

29 Simplify $(y^3)^4$.

30 Simplify $2x^3 \times 5x^2$.

Inequalities

- 31 Solve $3x - 4 > 11$.
- 32 Solve $-2x + 6 \leq 20$.
- 33 Solve $5x + 1 \geq 3x + 9$.
- 34 Solve $4(x - 2) < 2x + 6$.
-

Substitution into formulas

- 35 The formula for simple interest is $I = Prt$. Find I when $P = 2,000$, $r = 0.04$, $t = 3$.
- 36 The perimeter of a rectangle is $P = 2(l + w)$. Find P when $l = 15$ and $w = 9$.
- 37 If $v = u + at$, find v when $u = 5$, $a = 3$, $t = 4$.
- 38 The area of a triangle is $A = \frac{1}{2}bh$. Find A when $b = 14$ and $h = 9$.
-

Translating words into equations

- 39 Twice a number, plus 7, equals 25. Find the number.
- 40 A number decreased by 8 is equal to three times the number, decreased by 20. Find the number.
- 41 The sum of three consecutive integers is 84. Find the integers.
- 42 Five more than three times a number is the same as 10 less than five times the number. Find the number.
-

Word problems

- 43 A rectangle has length $(x + 3)$ cm and width x cm. This question has two parts.
- a If the area is 40 cm^2 , write an equation in x .
- b Solve for x .
- 44 A father is 3 times as old as his son. In 12 years, he will be twice as old as his son. Find their current ages.
- 45 Tickets cost 25 riyals for adults and 15 riyals for children. A group of 10 people paid 220 riyals in total. How many adults were in the group?

46 The sum of two numbers is 45, and their difference is 11. Find the two numbers.